

Lecture 10 : Differentiation

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Theorem 1 *Mean value theorem (Lagrange)*

Let a function $f : [a, b] \rightarrow \mathbb{R}$ be such that

1. f is continuous on $[a, b]$, and
2. f is differentiable at every point of (a, b) .

Then there exists at least a point ξ in (a, b) , such that

$$\frac{f(b) - f(a)}{b - a} = f'(\xi).$$

Proof: Let us define a function $\phi : [a, b] \rightarrow \mathbb{R}$ by $\phi(x) = f(x) + Ax$, $x \in [a, b]$, where A is a constant.

Clearly, ϕ is continuous on $[a, b]$, since f is continuous on $[a, b]$; and ϕ is differentiable at every point of (a, b) , since f is differentiable at every point of (a, b) .

Let us choose A , such that $\phi(a) = \phi(b)$.

Then, $f(a) + Aa = f(b) + Ab$, and this determines $A = \frac{f(b) - f(a)}{a - b}$. For this choice of A , ϕ satisfies all conditions of Rolle's theorem on $[a, b]$. Therefore, there exists at least a point ξ in (a, b) such that $\phi'(\xi) = 0$.

But $\phi'(\xi) = f'(\xi) + A$ and therefore $0 = f'(\xi) + \frac{f(b) - f(a)}{a - b}$, or $f'(\xi) = \frac{f(b) - f(a)}{b - a}$.

Example : Determine the number c , which satisfy the conclusion of Cauchy's mean value theorem for the following function

$$f(x) = x^3 + 2x^2 - x, \quad x \in [-1, 2].$$

Since, $x^3 + 2x^2 - x$ is continuous in the interval $[-1, 2]$. Also, $f(x)$ is differentiable in $(-1, 2)$ i.e

$$f'(x) = 3x^2 + 4x - 1.$$

Then, by Cauchy's mean value theorem there must be at least one $c \in (-1, 2)$, such that $f'(c) = \frac{f(2) - f(-1)}{2 - (-1)}$.

$$\implies f'(c) = 3c^2 + 4c - 1 = \frac{14 - 2}{2 + 1} = \frac{12}{3} = 4$$

$$\implies 3c^2 + 4c - 1 = 4$$

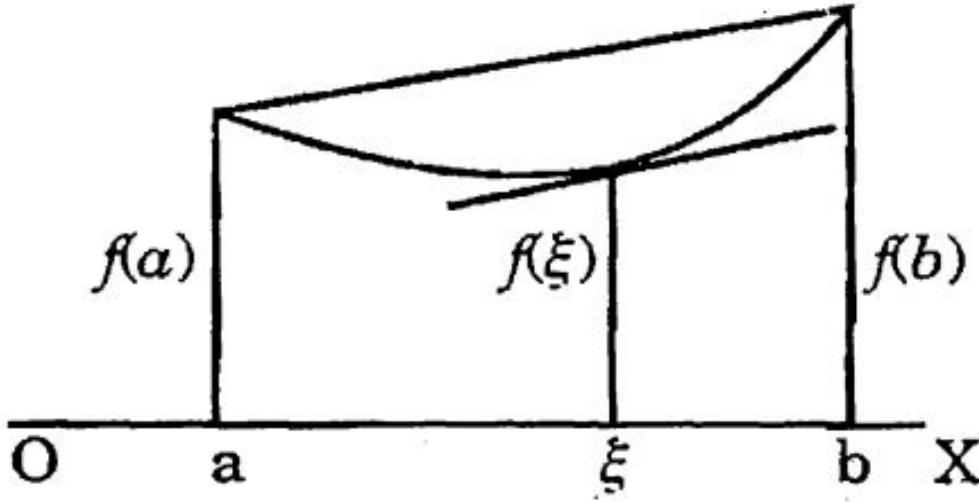
$$\implies 3c^2 + 4c - 5 = 0$$

$$\implies c = \frac{-4 + \sqrt{76}}{6} = 0.7863 \text{ or } c = \frac{-4 - \sqrt{76}}{6} = -2.1196.$$

Since, $0.7863 \in (-1, 2)$, therefore $c = 0.7863$.

Geometrical interpretation.

If a function f has a graph which is a continuous curve on the interval $[a, b]$ and the curve has a tangent at every point on it with abscissa between a and b , then there exists a point ξ in (a, b) , such that the tangent to the curve at $(\xi, f(\xi))$, is parallel to the line segment joining the points $(a, f(a))$ and $(b, f(b))$.



Theorem 2 Mean value theorem (Cauchy) Let the functions $f : [a, b] \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow \mathbb{R}$ be such that

1. f and g are both continuous on $[a, b]$,
2. f and g are both differentiable on (a, b) , and
3. $g'(x) \neq 0$, for all $x \in (a, b)$. Then, there exists a point ξ in (a, b) , such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}$$

Proof: Let $g(a) \neq g(b)$, Because, if $g(a) = g(b)$, then g would satisfy all conditions of Rolle's theorem on $[a, b]$ and consequently $g'(x)$ would be equal to 0, for some point c in (a, b) and this would contradict the condition (3), of the theorem.

Let us define, $\phi : [a, b] \rightarrow \mathbb{R}$ by $\phi(x) = f(x) + Ag(x)$, $x \in [a, b]$, where A is a constant. ϕ is continuous on $[a, b]$, since f and g are both continuous on $[a, b]$; ϕ is differentiable on (a, b) , since f and g are both differentiable on (a, b) .

Let us choose A $f(a) + Ag(a) = f(b) + Ag(b)$ or, $A[g(a) - g(b)] = f(b) - f(a)$. Since,

$$g(a) \neq g(b), \quad A = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

For this choice of A , ϕ satisfies all conditions of Rolle's theorem on $[a, b]$. Therefore, there exists a point ξ in (a, b) , such that $\phi'(\xi) = 0$.

But,

$$\phi'(\xi) = f'(\xi) + \frac{f(b) - f(a)}{g(a) - g(b)} g'(\xi).$$

As $a < \xi < b$, $g'(\xi) \neq 0$ and therefore

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}.$$

This completes the proof.

Reference book : *Introduction to real analysis by S.K. Mapa.*